

Evaluating Pre-Trained Vision Transformers Under Continual Learning

Master's Degree in Computer Science

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SAPIENZA
UNIVERSITÀ DI ROMA



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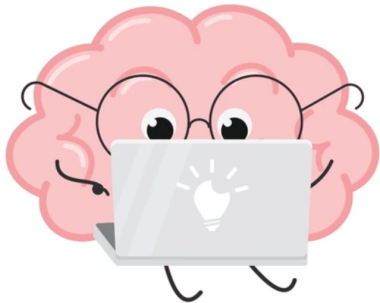
► Results and Interpretation



Human Learning Vs Machine Learning

1 Motivation and Research Objective

Humans learn **continuously**, while machines often forget what they learned before.





Main Research Question

1 Motivation and Research Objective

Can pre-trained ViTs, combined with **parameter-efficient adaptation and regularization**, serve as effective Continual Learning systems?



Key Contributions

1 Motivation and Research Objective

- **Systematic evaluation of pre-trained ViT-Tiny** under a task-incremental CL framework, compared to a CNN baseline.



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- **Systematic evaluation of pre-trained ViT-Tiny** under a task-incremental CL framework, compared to a CNN baseline.
- **Parameter-efficient adaptation using bottleneck adapters**, studying how freezing the backbone and limiting updates affects retention.



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- **Systematic evaluation of pre-trained ViT-Tiny** under a task-incremental CL framework, compared to a CNN baseline.
- **Parameter-efficient adaptation using bottleneck adapters**, studying how freezing the backbone and limiting updates affects retention.
- **Combining adapter modules with EWC regularization**, applying EWC only to adapter parameters to improve the stability-plasticity balance.



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- **Systematic evaluation of pre-trained ViT-Tiny** under a task-incremental CL framework, compared to a CNN baseline.
- **Parameter-efficient adaptation using bottleneck adapters**, studying how freezing the backbone and limiting updates affects retention.
- **Combining adapter modules with EWC regularization**, applying EWC only to adapter parameters to improve the stability-plasticity balance.
- **Structured ablation study across four configurations**, isolating each component's contribution to CL performance.



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Core Components of the Proposed Approach

2 Methodology and Experimental Setup

Adapters create a localized trainable parameter space, making EWC more focused and precise.



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ViT-Tiny

12 transformer blocks,
192-dimensional
embeddings, and
3 attention heads.

Pre-trained on ImageNet-1K
and processes images as
patch tokens.



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EWC Regularization

Uses Fisher Information to
identify important adapter
weights.
Large changes to these
weights are penalized.



Four-Stage Ablation Study

2 Methodology and Experimental Setup

ResNet-18

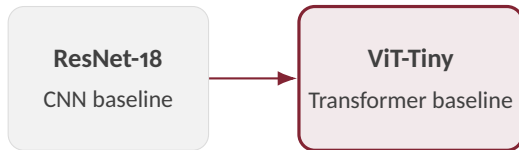
CNN baseline

Stage 1: CNN baseline trained sequentially without any CL-specific strategy.



Four-Stage Ablation Study

2 Methodology and Experimental Setup

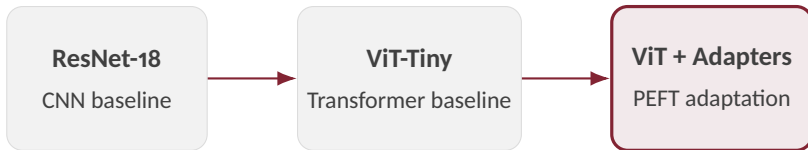


Stage 2: ViT baseline trained sequentially without any CL-specific strategy.



Four-Stage Ablation Study

2 Methodology and Experimental Setup

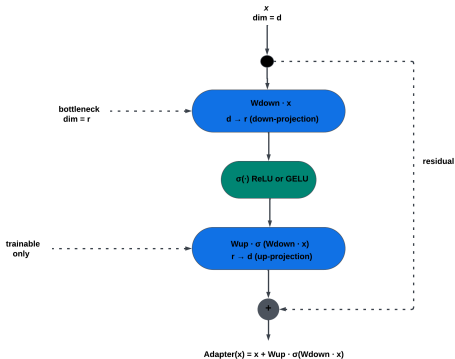


Stage 3: Adapter modules are introduced to enable parameter-efficient adaptation.



Adapter Module Architecture

2 Methodology and Experimental Setup



Bottleneck Design

The adapter compresses the representation into a smaller hidden dimension.

It then projects it back and adds it through a residual connection.

Structure: $192 \rightarrow 64 \rightarrow 192$



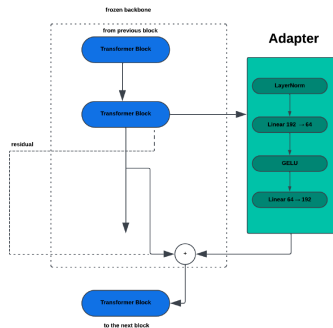
Adapter Placement in the ViT Backbone

2 Methodology and Experimental Setup

Placement

Adapters are inserted after each transformer block.

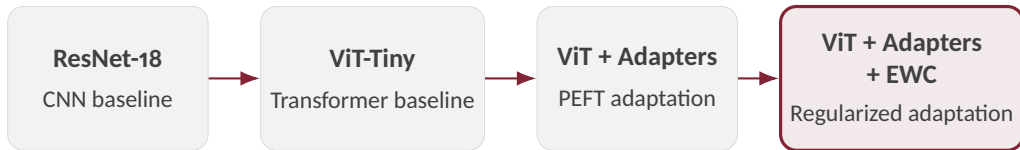
The ViT backbone remains frozen, while only adapter parameters are updated.





Four-Stage Ablation Study

2 Methodology and Experimental Setup



Stage 4: EWC is applied only to adapter parameters to protect important weights across tasks.



Experimental Results

2 Methodology and Experimental Setup

Model	Avg. Accuracy	Avg. Forgetting	BWT
ResNet-18	24.22%	69.17%	-69.17%
ViT-Tiny	41.91%	63.26%	-63.26%
ViT + Adapters	76.93%	19.91%	-19.91%
ViT + Adapters + EWC	90.59%	2.01%	-1.85%



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ViT + Adapters + EWC	90.59%	2.01%	-1.85%

Performance Trend

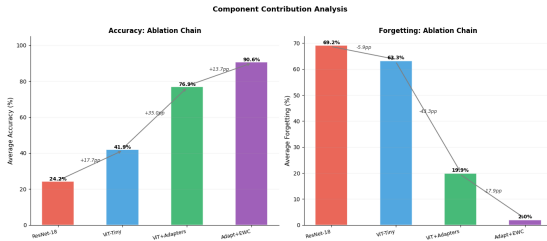




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Why Adapters and EWC Reduce Forgetting?

3 Results and Interpretation

Frozen Backbone

Pre-trained ViT features remain fixed, so shared representations are not overwritten.



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Adapter Restriction

Only 5.5% of parameters are updated, reducing interference between tasks.



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Important adapter weights are protected using Fisher-based regularization.



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Only 5.5% of parameters are updated, reducing interference between tasks.

EWC on Adapters

Important adapter weights are protected using Fisher-based regularization.

Per-task forgetting with EWC: 2.85%, 0%, 1.30%, 3.90%. No monotonic pattern appears, suggesting Fisher-based protection rather than task-order bias.



Robustness Check Across Random Seeds

3 Results and Interpretation

Mean Accuracy

86.93%

$\pm 1.78\%$

Mean Forgetting

5.91%

$\pm 1.61\%$

Accuracy Gain

+13.66pp

vs. adapters-only

Forgetting Drop

-17.9pp

vs. adapters-only

Per-seed breakdown

Run	Seed	Avg. Accuracy	Avg. Forgetting	BWT
1	42	86.19%	5.51%	-5.51%
2	123	89.39%	4.18%	-4.18%
3	7	85.22%	8.05%	-8.05%

Key takeaway: The improvement remains consistent across different task orderings and class partitions.



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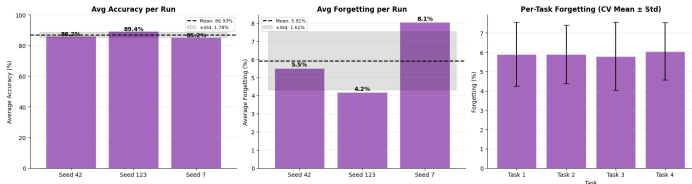
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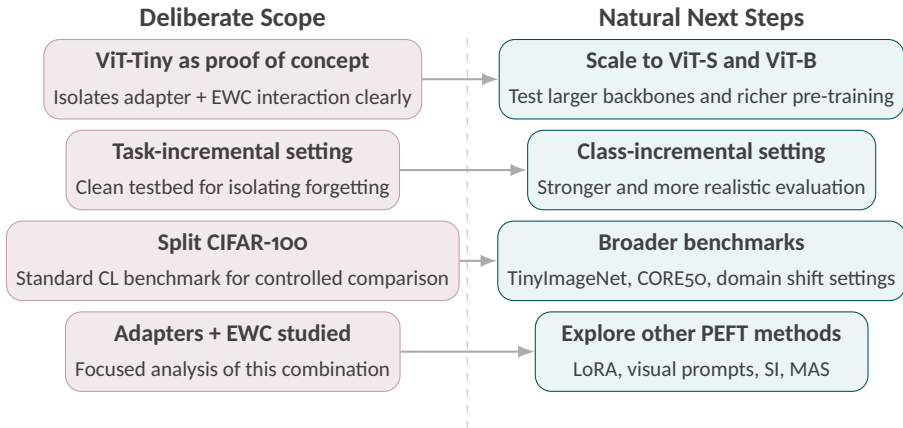


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Scope and Future Directions

3 Results and Interpretation



The current setup is intentionally controlled. The next step is to test whether the same adapter-EWC interaction scales to harder continual learning settings.



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Thank you for listening!
Any questions?